

RepoMedic

AI-Powered Repository Health & Debugging Assistant

Scan → Analyze → Explain → Fix → Verify

IBM Bob × Python × AI

Developer workflow

A practical workflow for understanding unfamiliar repositories, tracing issues, estimating impact, applying bounded fixes, and checking the result.

Hackathon Project

The problem

Debugging a complex repository is often a chain of manual investigation.

1

Something breaks

The first error does not always reveal the real cause.

2

Find the right file

Developers search across code, configuration and dependencies.

3

Trace the impact

A small change can affect several connected modules.

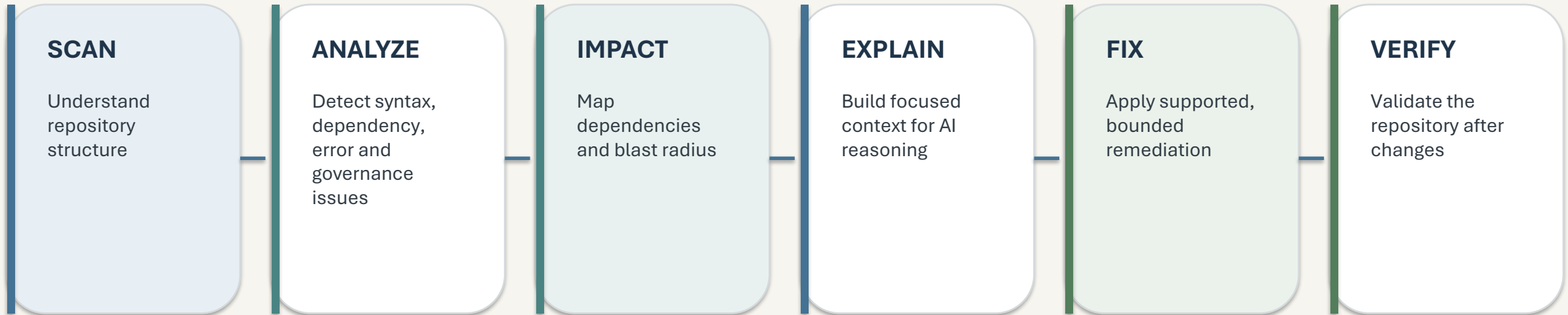
4

Fix and retest

A fix is only useful if the repository remains healthy.

The solution

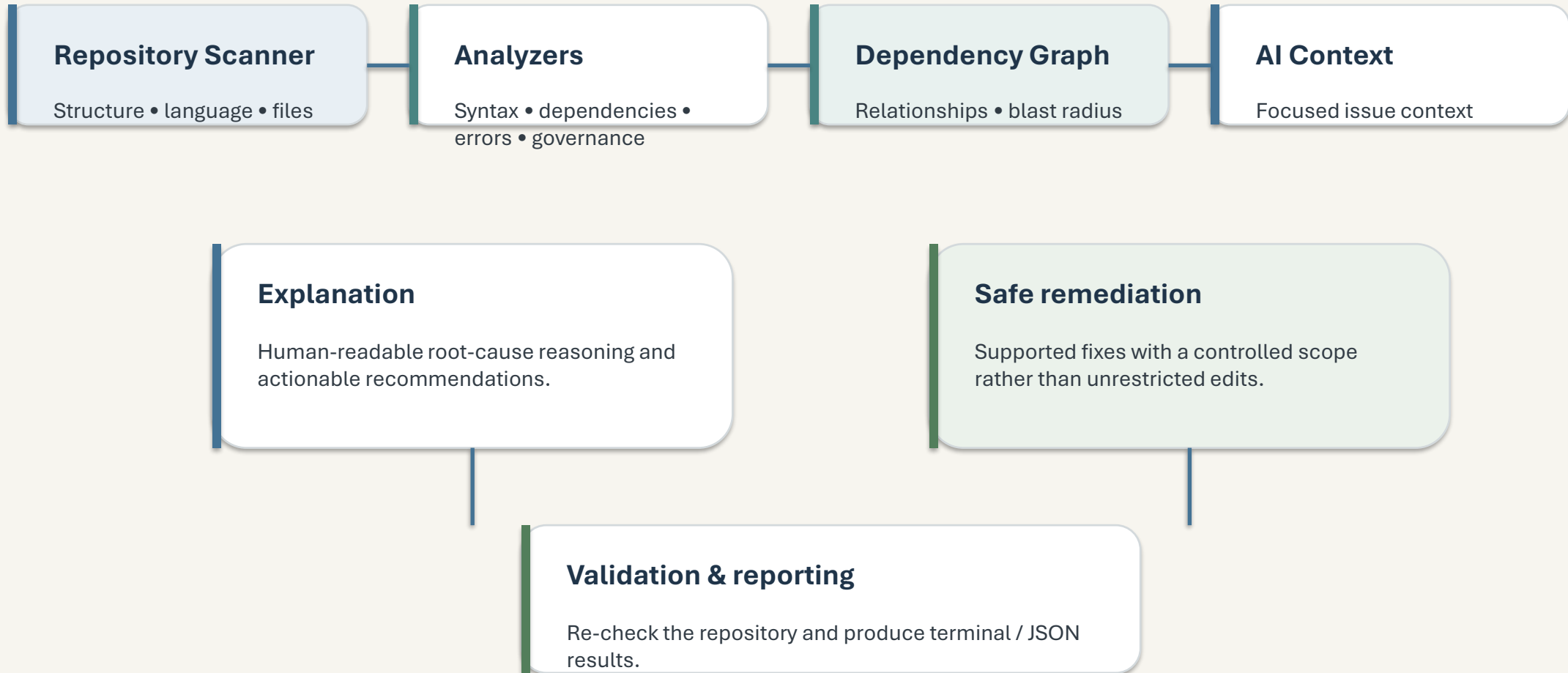
RepoMedic puts repository diagnosis and verification into one repeatable flow.



AI assists reasoning, while deterministic tooling supplies repository facts and verification.

How RepoMedic works

A modular architecture separates facts, reasoning, remediation and validation.



What makes the workflow useful

The system focuses on context, impact and verification—not just raw error messages.

Repository intelligence

Detects project structure, technologies and relevant repository facts.

Multi-layer analysis

Combines syntax, dependency, error and governance checks.

Dependency & blast radius

Shows which connected modules may be affected by a problem or change.

AI root-cause explanation

Turns findings and focused context into an understandable explanation.

Safe remediation

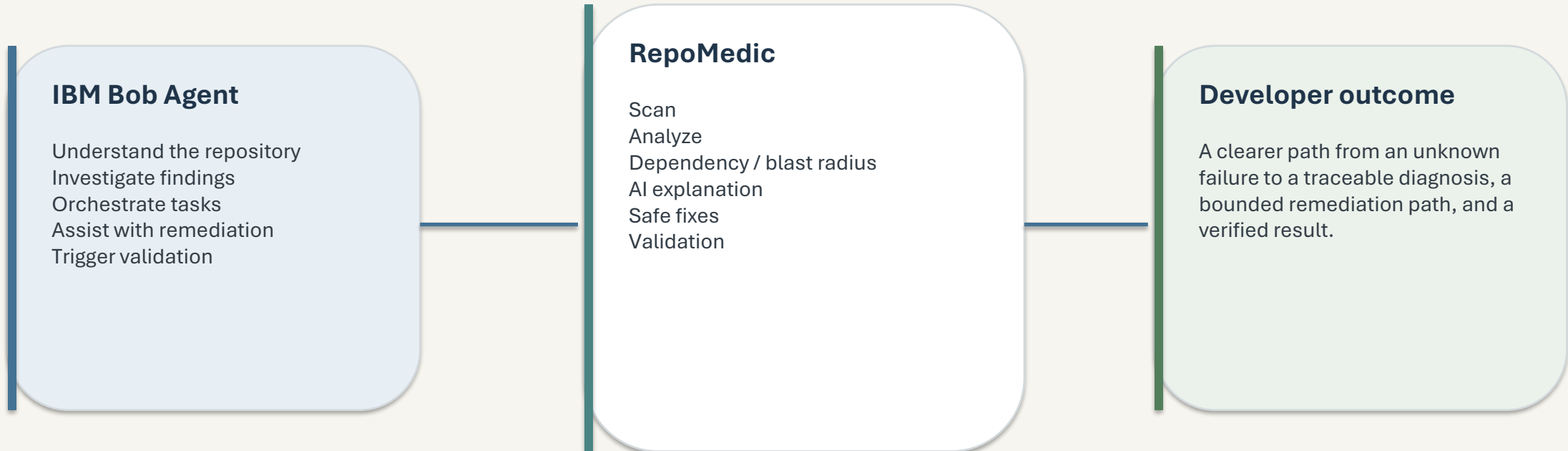
Keeps automated changes bounded and explicit.

Verification

Checks the repository after remediation instead of assuming success.

IBM Bob in the workflow

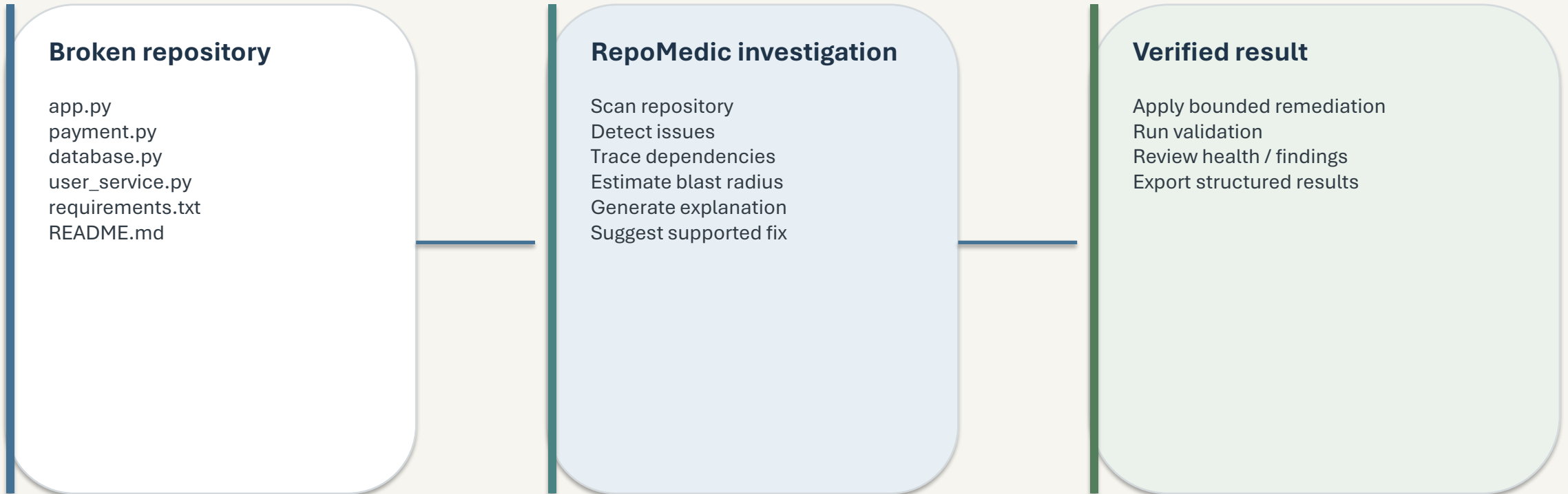
Bob is the AI-assisted development and orchestration layer around the repository workflow.



During development, IBM Bob was also used to accelerate implementation, debugging, testing, refactoring and iteration.

Demo scenario

A deliberately broken application makes the workflow easy to see.



“Something is broken” → “Here is what happened, what it affects, what can be done, and whether the fix worked.”

RepoMedic

From guessing to a traceable debugging workflow.

Scan → Analyze → Explain → Fix → Verify

Built for developers

Python CLI • Repository analysis • Dependency mapping • AI-assisted explanation • Safe remediation • Validation

Thank you